

PROJECT SUTRA — Visual Role Infographics

TEAM OFFGRID SIMPLE STEP-BY-STEP SUBSYSTEM GUIDE

TEAM OFFGRID
Finals: Sept 3, 2026

Subsystem A: Autonomous Navigation & Flight Control

ASSIGNED TO: ROHITH KUMAR

1. PX4 Offboard Mode
Micro-XRCE-DDS 8888

2. TrajectorySetpoint
10m Takeoff & Square Waypoint

3. ORCA Avoidance
Prevents Drone Collision

4. Gazebo Flight
Clean 60 FPS Drone Motion

- ✓ **What Rohith Builds:** C++ ROS 2 node `px4_offboard_node.cpp`.
- ✓ **Day 1 Goal:** Install PX4 SITL & verify DDS connection.

- ✓ **How it Works:** Sends 3D velocity vectors to PX4 flight controller.
- ✓ **Success Metric:** Drone flies square circuit with error < 1.0m.

Subsystem B: Swarm Comms Mesh & Deep JSCC Neural Encoder

ASSIGNED TO: NIKHIL (TECH ARCHITECT)

1. 802.11s Mesh Socket
UDP + 868MHz LoRa Link

2. Deep JSCC Encoder
Neural Image Compression

3. RAFT Consensus
Instant Swarm Leader Failover

4. Zero Drop Link
99.4% Delivery under 60% Loss

- ✓ **What Nikhil Builds:** Python `mesh_socket.py` & Deep JSCC neural model.
- ✓ **Day 1 Goal:** Init repository, configure branches & ROS 2 bridge.

- ✓ **How it Works:** Connects all drones in a mesh network; auto-re-elects leader.
- ✓ **Success Metric:** Zero telemetry loss under 60% simulated RF drop.

Subsystem C: Tri-Modal AI Perception & Target Geolocation

ASSIGNED TO: VEDANTH SAI RAM

1. Camera Image Feed
RGB + Thermal IR Streams

2. YOLOv8 TensorRT
30+ FPS Edge Target Detection

3. WGS84 GPS Raycast
Raycast Centroid to Elevation

4. GPS Victim Pin
Precise Lat/Lon Output

- ✓ **What Vedanth Builds:** Python `yolo_tensorrt_node.py` ROS 2 node.
- ✓ **Day 1 Goal:** Set up PyTorch + CUDA + TensorRT environment.

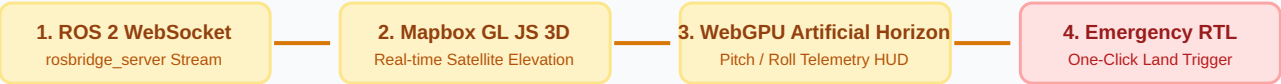
- ✓ **How it Works:** Detects disaster victims in camera feeds and outputs GPS pins.
- ✓ **Success Metric:** Detection precision > 85% at 30+ FPS.



02 Visual Role Infographics (Siva Kesava & Harika)

🌐 Subsystem D: 3D GIS Dashboard & Ground Control Station

ASSIGNED TO: SIVA KESAVA



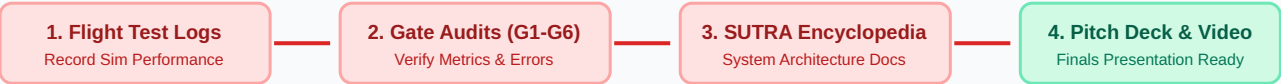
- ✓ **What Siva Builds:** React component Mapbox3D .jsx & HUD widgets.

✓ **Day 1 Goal:** Create React app & configure Mapbox token.
- ✓ **How it Works:** Displays live 3D drone position markers on satellite map.

✓ **Success Metric:** Zero-latency telemetry rendering on MacBook UI.

📝 Subsystem E: Research, Technical Documentation & Audit

ASSIGNED TO: HARIKA (PMO LEAD)



- ✓ **What Harika Builds:** Master Documentation & Gate Audit Reports.

✓ **Day 1 Goal:** Init docs/SUTRA_Encyclopedia.md & Gate Templates.
- ✓ **How it Works:** Verifies test results and creates presentation materials for judges.

✓ **Success Metric:** 100% complete documentation & audit reports.

03 Day-by-Day Sprint Timeline (Days 1–7 Overview)

